

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 13

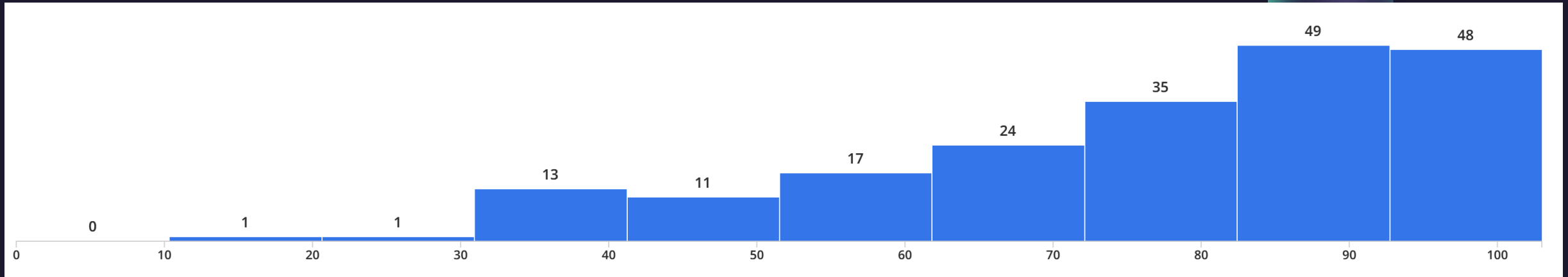


Any Reminders? Any Questions?

- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!
- Quiz on Friday on polar coordinates and beginning of triple integrals

EXAM 1

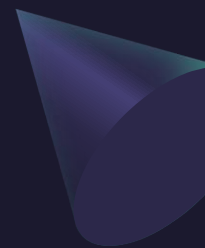
- Average = 77, Median = 81



- Exam I – One-Page Correction (+5)



ALVARO: Start the recording!



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More on Triple Integrals

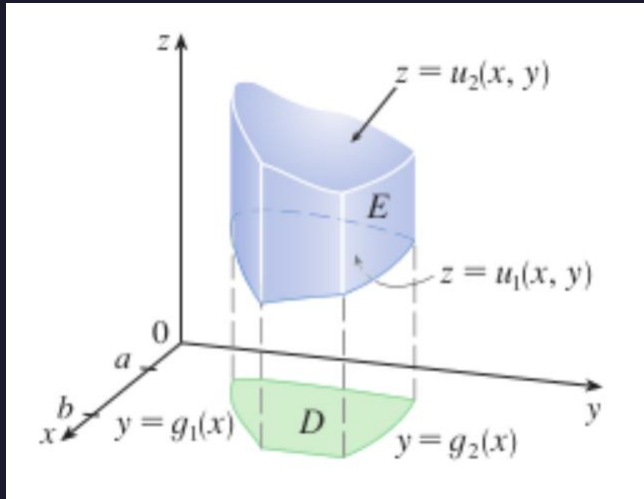




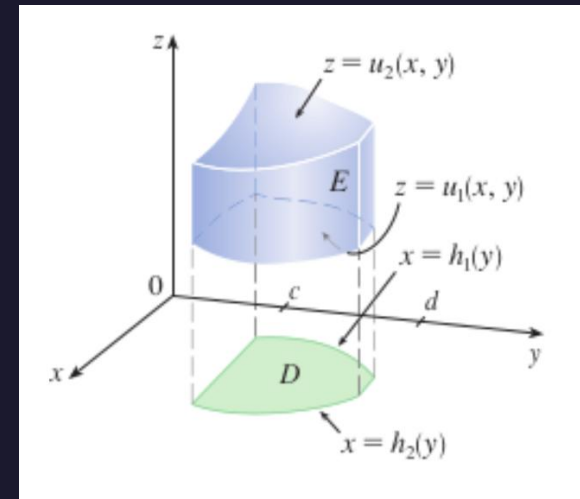
Today – Triple Integrals

- Triple Integrals in Boxes
- Triple Integrals in Regions
- Changing the Order of Integration

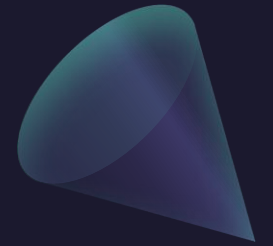
Triple Integrals in General Regions: Type I, ...



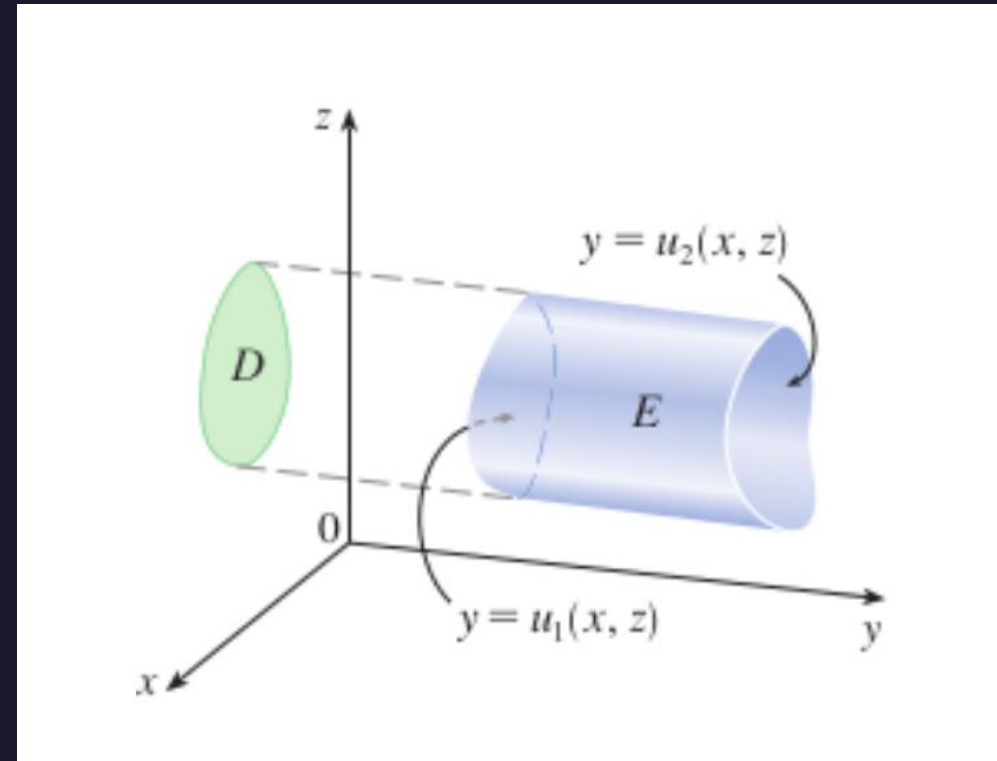
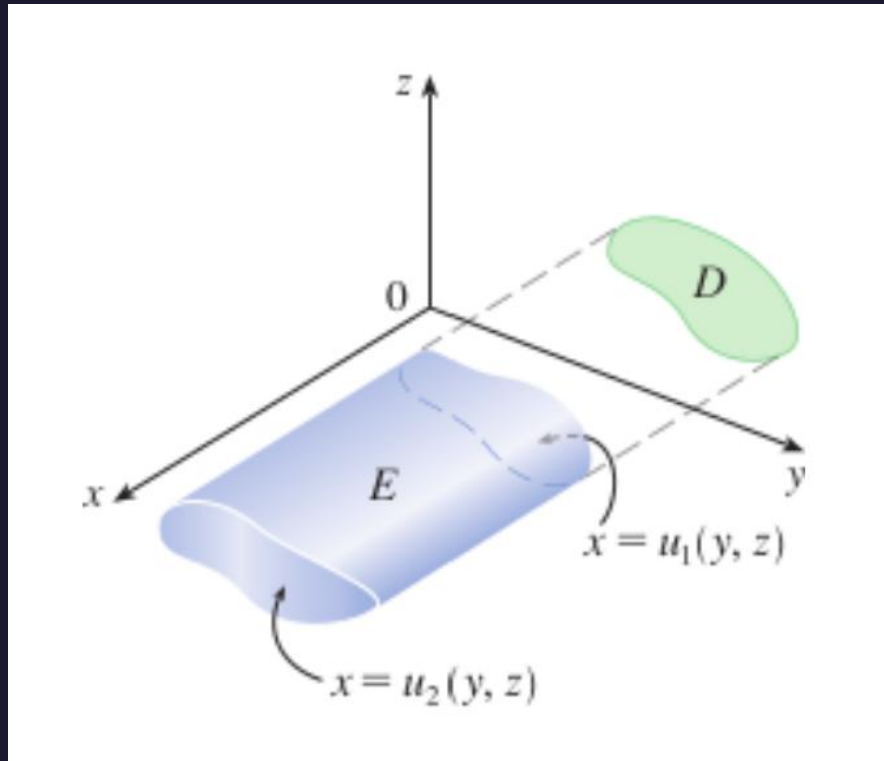
$$\iiint_E f(x, y, z) dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz dx dy$$



$$\iiint_E f(x, y, z) dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz dx dy$$



Triple Integrals in General Regions: Type II and Type III



Example: Evaluate the following triple integral

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Example: Evaluate the following triple integral

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

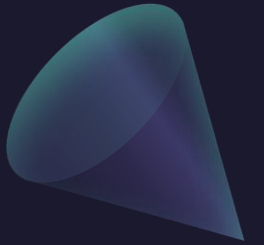
$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Changing the Order of Integration

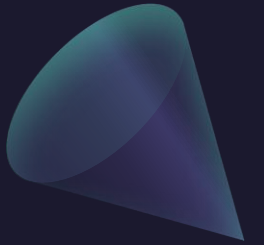
Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

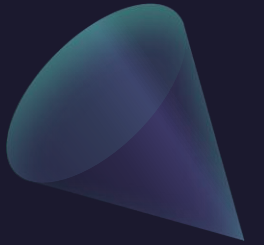
Example: Find the volume of the solid bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$.



Example: Find the volume of the solid bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$.

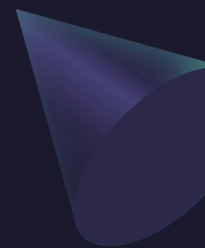


Questions?





ALVARO: Start the recording!

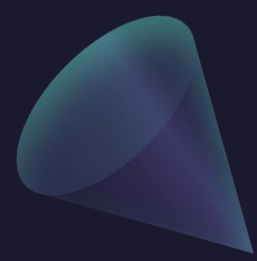
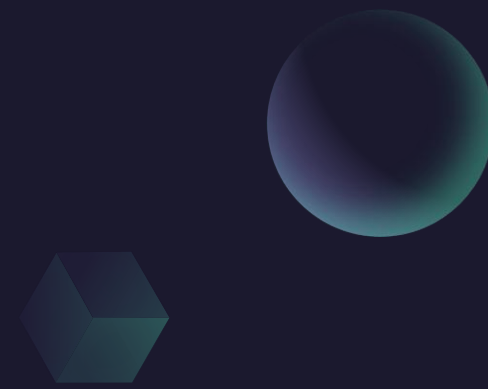


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Multi-Variable Calculus

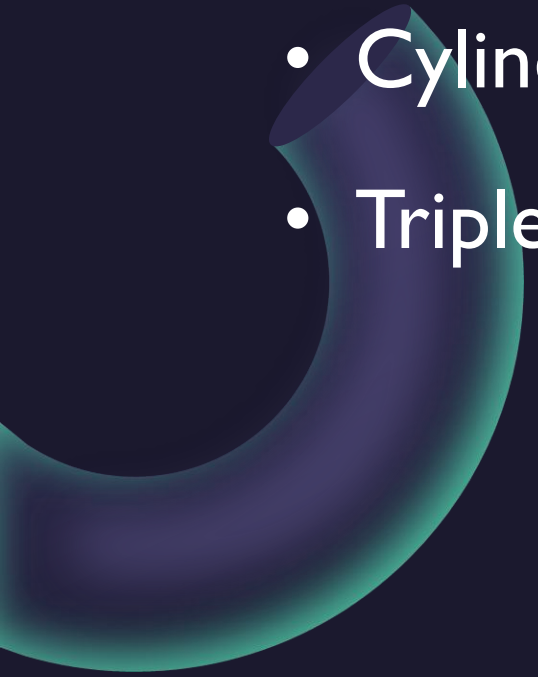
Instructor: Álvaro Lozano-Robledo

Triple Integrals in Cylindrical



A small, dark teal sphere is positioned above a small, dark teal cube, both located to the left of the main title.

Today – Triple Integrals in Cylindrical Coordinates

- 
- A large, dark teal torus (donut shape) is positioned on the left side of the slide, partially overlapping the list of topics.
- Cylindrical Coordinates
 - Triple Integrals in Cylindrical Coordinates

Cylindrical Coordinates

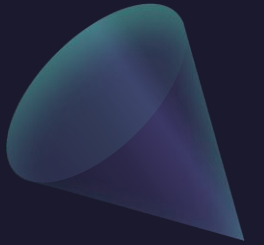
$$x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

$$r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x} \quad z = z$$

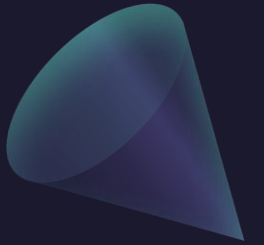
Example: Plot the point with cylindrical coordinates $(r, \theta, z) = (2, \pi, 3)$

Example: Find the cylindrical coordinates of the points $(1,0,0)$, $(0,0,1)$, and $(1,1,1)$.

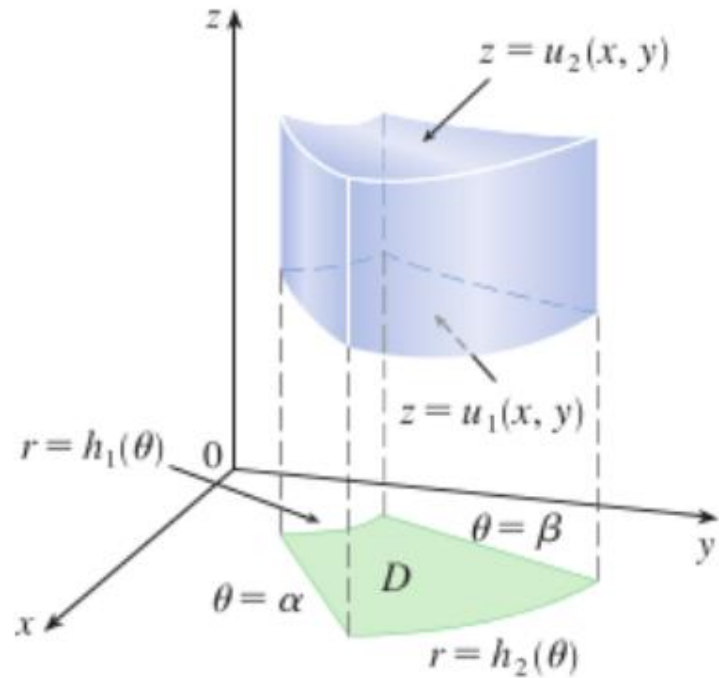
Example: Describe the surface given in cylindrical coordinates by $z = r$.



Example: Describe the surface given in cylindrical coordinates by $z = r^2$.



Triple Integrals in Cylindrical Coordinates (of Type I)



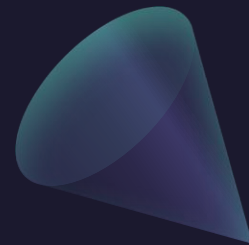
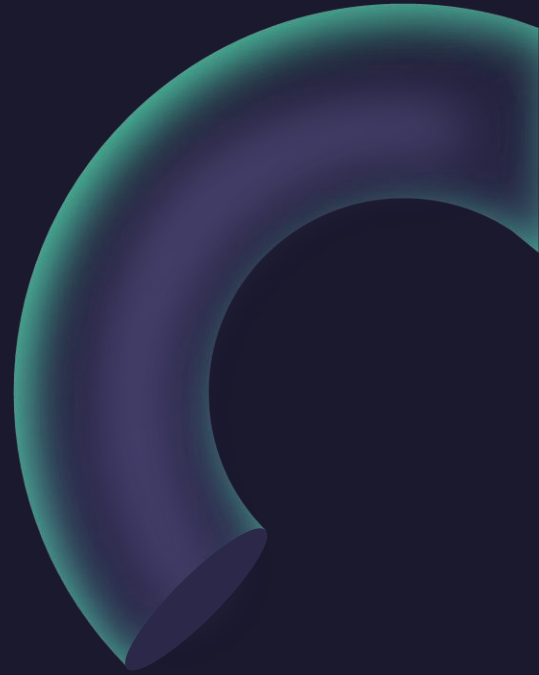
$$\iiint_E f(x, y, z) \, dV = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \right] dA$$

$$\iiint_E f(x, y, z) \, dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} f(r \cos \theta, r \sin \theta, z) \, r \, dz \, dr \, d\theta$$

Example: Evaluate the value of the triple integral

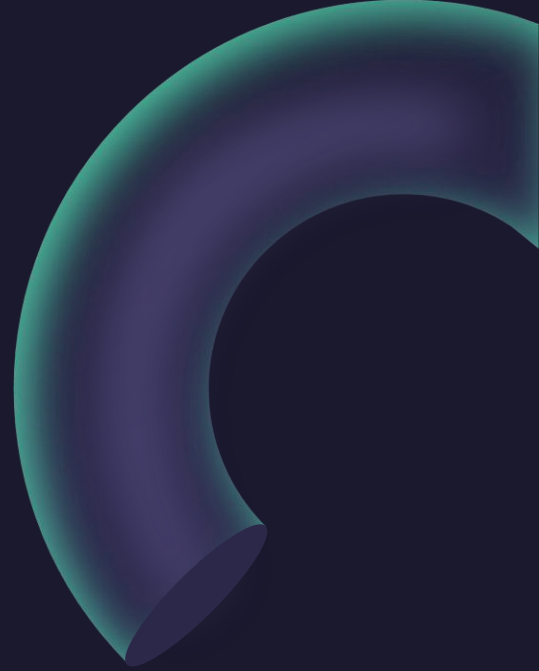
$$\iiint_E x^2 dV$$

where E is the solid bounded by the surface
 $z = 4 - x^2 - y^2$ and above the xy -plane.



Example: Evaluate the value of the triple integral

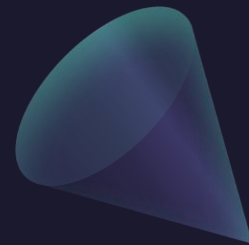
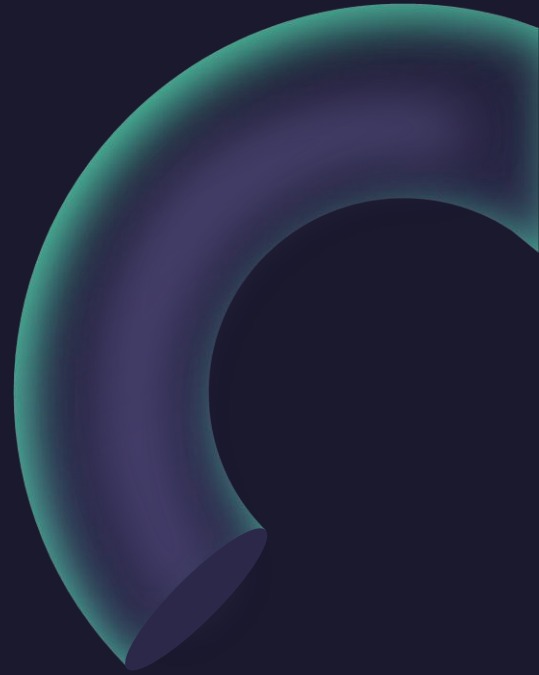
$$\iiint_E x^2 dV$$



$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$$

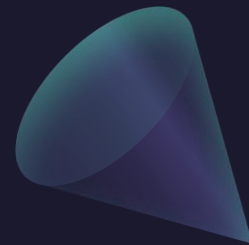
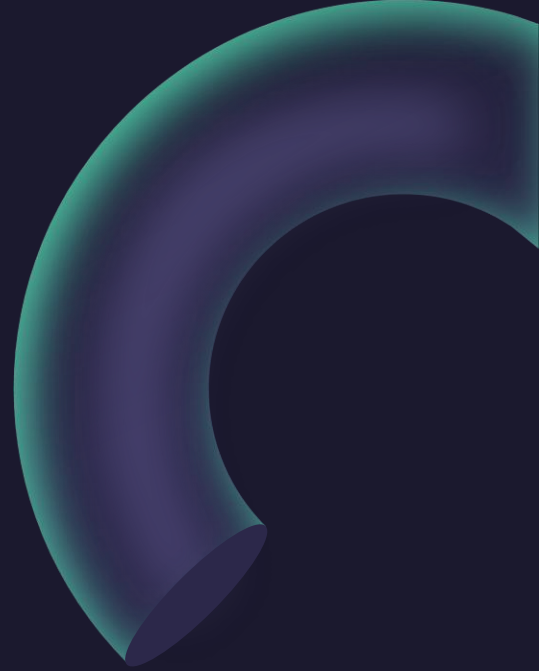
Example: Evaluate the value of the triple integral

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) \, dz \, dy \, dx$$



Example: Evaluate the value of the triple integral

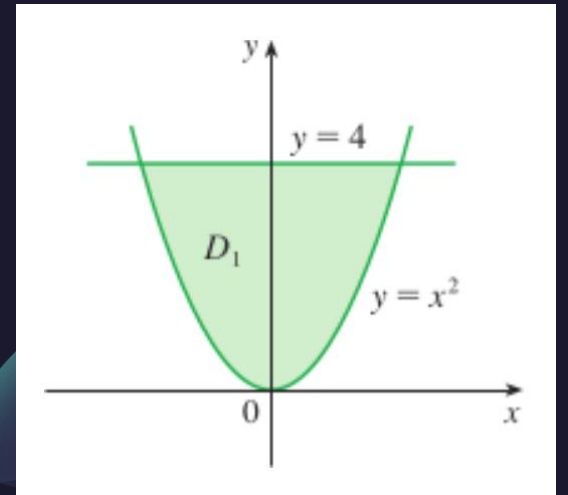
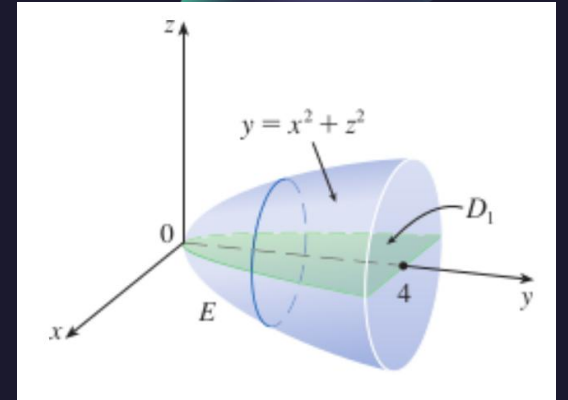
$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) \, dz \, dy \, dx$$



Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + z^2} \, dV$$

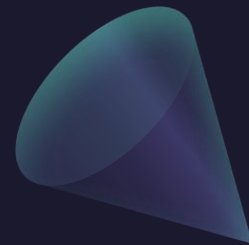
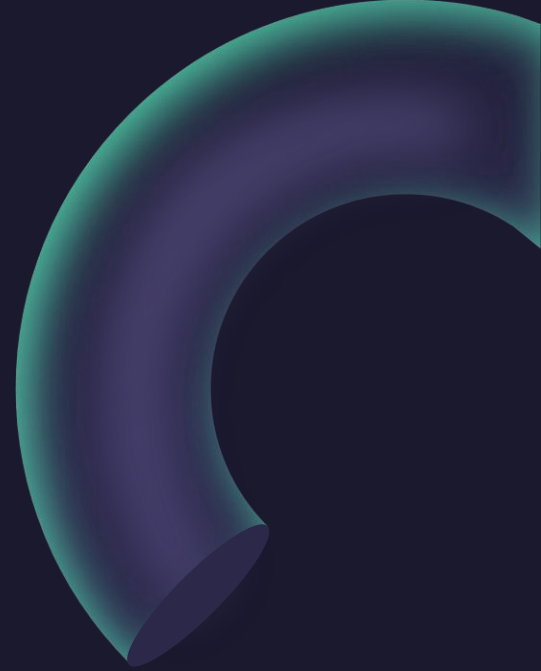
where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.



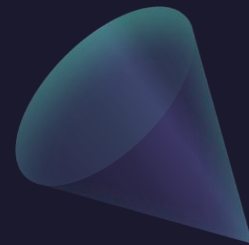
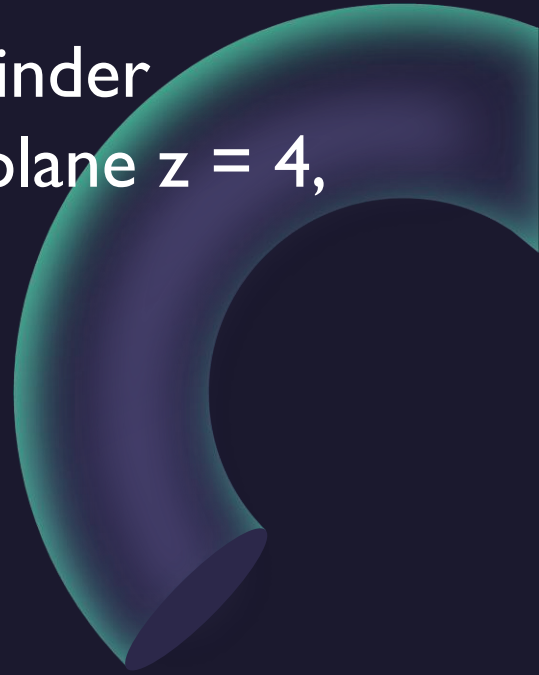
Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + z^2} \, dV$$

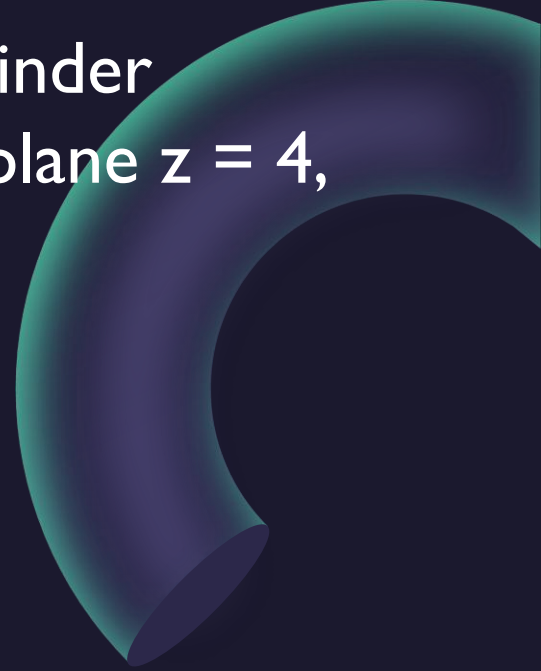
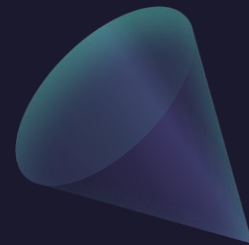
where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.



Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane $z = 4$, and above $z = 1 - x^2 - y^2$.



Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane $z = 4$, and above $z = 1 - x^2 - y^2$.



Thank you

Until next time.

